

Operating Systems
Lab 9

1. Write a C program using the pthread's semaphore to implement the Bounded Buffer problem. You must implement pseudo-code for the producer as the "insertbuffer()" function, which will be executed by a producer thread created by the main program. However, instead of using a forever loop, implement the "insertbuffer()" to run only forty rounds. The function shows the producer thread's work, as is shown in the example output. The pseudo-code for consumer must be implemented as the "readbuffer()" function where the consumer thread executes this function, the forever loop needs to be changed to run only forty rounds. The function shows the consumer thread's work, as is shown in the example output. The main program initializes all semaphores, creates producer and consumer threads, waits for all threads to finish, and destroys all semaphores. The example output of the program is as follows:

```
Producer Entered 0 Buffer size = 1
Producer Entered 1 Buffer size = 2
Producer Entered 2 Buffer size = 3
Producer Entered 3 Buffer size = 4
Producer Entered 4 Buffer Full
Consumer consumed 0 buffer size = 4
Consumer consumed 1 buffer size = 3
Consumer consumed 2 buffer size = 2
Consumer consumed 3 buffer size = 1
Consumer consumed 4 Buffer Empty
Producer Entered 5 Buffer size = 1
Producer Entered 6 Buffer size = 2
Producer Entered 7 Buffer size = 3
Producer Entered 8 Buffer size = 4
Producer Entered 9 Buffer Full
Consumer consumed 5 buffer size = 4
Consumer consumed 6 buffer size = 3
Consumer consumed 7 buffer size = 2
Consumer consumed 8 buffer size = 1
Consumer consumed 9 Buffer Empty
Producer Entered 10 Buffer size = 1
Producer Entered 11 Buffer size = 2
Producer Entered 12 Buffer size = 3
Producer Entered 13 Buffer size = 4
Producer Entered 14 Buffer Full
Consumer consumed 10 buffer size = 4
Consumer consumed 11 buffer size = 3
Consumer consumed 12 buffer size = 2
Consumer consumed 13 buffer size = 1
Consumer consumed 14 Buffer Empty
Producer Entered 15 Buffer size = 1
Producer Entered 16 Buffer size = 2
Producer Entered 17 Buffer size = 3
Producer Entered 18 Buffer size = 4
Producer Entered 19 Buffer Full
Consumer consumed 15 buffer size = 4
Consumer consumed 16 buffer size = 3
Consumer consumed 17 buffer size = 2
Consumer consumed 18 buffer size = 1
Consumer consumed 19 Buffer Empty
Producer Entered 20 Buffer size = 1
Producer Entered 21 Buffer size = 2
Producer Entered 22 Buffer size = 3
Producer Entered 23 Buffer size = 4
Producer Entered 24 Buffer Full
Consumer consumed 20 buffer size = 4
Consumer consumed 21 buffer size = 3
Consumer consumed 22 buffer size = 2
Consumer consumed 23 buffer size = 1
Consumer consumed 24 Buffer Empty
Producer Entered 25 Buffer size = 1
Producer Entered 26 Buffer size = 2
Producer Entered 27 Buffer size = 3
Producer Entered 28 Buffer size = 4
Producer Entered 29 Buffer Full
Consumer consumed 25 buffer size = 4
Consumer consumed 26 buffer size = 3
Consumer consumed 27 buffer size = 2
Consumer consumed 28 buffer size = 1
Consumer consumed 29 Buffer Empty
Producer Entered 30 Buffer size = 1
Producer Entered 31 Buffer size = 2
Producer Entered 32 Buffer size = 3
Producer Entered 33 Buffer size = 4
Producer Entered 34 Buffer Full
Consumer consumed 30 buffer size = 4
Consumer consumed 31 buffer size = 3
Consumer consumed 32 buffer size = 2
Consumer consumed 33 buffer size = 1
Consumer consumed 34 Buffer Empty
Producer Entered 35 Buffer size = 1
Producer Entered 36 Buffer size = 2
Producer Entered 37 Buffer size = 3
Producer Entered 38 Buffer size = 4
Producer Entered 39 Buffer Full
Consumer consumed 35 buffer size = 4
Consumer consumed 36 buffer size = 3
Consumer consumed 37 buffer size = 2
Consumer consumed 38 buffer size = 1
Consumer consumed 39 Buffer Empty
```